

Parameter-Efficient Adaptation of V-JEPA 2 for Egocentric Action Anticipation

Ayoub Sahlaoui¹, Salaheddin Alakkari², and Younes Dhassi¹

¹*Department of Computer Science, Sidi Mohamed Ben Abdellah University*

²*Department of Computer Science and Information Systems, University of Limerick*

Abstract

Egocentric action anticipation requires predicting an action before it begins, given only the preceding visual context. We test whether low-rank adaptation (LoRA) can parameter-efficiently specialize V-JEPA 2, a self-supervised video encoder trained with a feature-prediction objective. On a controlled 22-participant subset of EPIC-KITCHENS-100 with a shared attentive probe, we compare frozen probing against LoRA at ranks $r \in \{4, 8, 16, 32\}$ and a matched-budget upper-layer variant. Full-layer LoRA with $r = 4$ reaches the best action mean-class Recall@5, raising it from 16.39% to 18.67% with 0.39M trainable parameters (0.13% of the backbone). Higher ranks bring no further gain, and concentrating the same adapter budget in the upper six layers hurts performance. These results show that small, broadly distributed low-rank updates can adapt V-JEPA 2 without full fine-tuning, while trading extra training memory and time for a compact stored adapter.

Keywords: Action Anticipation, Parameter-Efficient Fine-Tuning, LoRA, V-JEPA, Egocentric Video

1 Introduction

Action anticipation asks a model to predict what a person will do before the action begins. Unlike recognition, the target action itself is unobserved. Egocentric video makes this harder: object interactions are fine-grained and class frequencies are heavily skewed. Anticipation matters for assistive systems and embodied agents that must act before an event has completed.

Large-scale self-supervised video encoders are a strong starting point for this problem. V-JEPA 2 [Assran et al., 2025] is a recent large-scale self-supervised video encoder pre-trained with a feature-prediction objective. Rather than reconstructing pixels, a joint-embedding predictive architecture (JEPA) learns to predict latent representations of masked content. JEPA models are a natural fit for anticipation, since both pre-training and downstream evaluation require reasoning about unobserved information.

Full fine-tuning of a large video backbone is expensive and risks overwriting useful pre-trained representations. Parameter-efficient fine-tuning (PEFT) sidesteps this by learning a small number of task-specific parameters while keeping the pre-trained weights fixed. Low-rank adaptation (LoRA) [Hu et al., 2022] is now a standard tool for adapting large foundation models, but it has been studied mainly on language models and reconstruction- or contrastively-trained visual encoders. Whether the low-rank update hypothesis behind LoRA holds for representations learned through feature prediction, as in a JEPA video model, has not been established empirically. To our knowledge, this is the first systematic evaluation of LoRA adaptation of V-JEPA 2 for egocentric action anticipation. We treat this as an empirical question rather than a claim of methodological novelty: our aim is a practical, evidence-based recipe for adapting V-JEPA 2.

Our study yields three findings:

- LoRA adaptation of V-JEPA 2 improves egocentric action anticipation over frozen probing, raising action mR@5 from 16.39% to 18.67% with only 0.39M adapter parameters (0.13% of the backbone).

- The downstream update has low intrinsic rank: sweeping $r \in \{4, 8, 16, 32\}$ shows that $r = 4$ is already optimal and higher ranks bring no gain while beginning to overfit.
- Where adapters are placed matters more than how large they are: under a fixed 0.39M budget, spreading small adapters across all 24 encoder layers beats concentrating them in the upper six layers by 1.30 pp.

2 Related Work

Egocentric action anticipation. EPIC-KITCHENS introduced large-scale egocentric video understanding in everyday kitchens [Damen et al., 2018], and EPIC-KITCHENS-100 expanded the dataset and evaluation protocol [Damen et al., 2022]. Action anticipation predicts a future verb–noun action before its onset. Prior work has explored attention-based modality fusion for this setting [Furnari and Farinella, 2019]. We focus on adapting a self-supervised video encoder under a controlled anticipation protocol.

Joint-embedding predictive architectures. JEPA models learn by predicting latent representations of masked content rather than reconstructing pixels [Assran et al., 2023]. V-JEPA 2 [Assran et al., 2025] scales this idea to internet-scale video and shows strong transfer to video understanding and anticipation; V-JEPA 2.1 later reports further gains through denser self-supervision [Mur-Labadia et al., 2026]. We study how much a JEPA video encoder can be specialized with few trainable parameters when the egocentric training set is small.

Video representation learning. Video encoders have benefited from supervised pre-training on action datasets such as Kinetics [Kay et al., 2017], contrastive learning [Chen et al., 2020, He et al., 2020, Radford et al., 2021], and masked reconstruction [He et al., 2022, Tong et al., 2022]. JEPA models differ from reconstruction-based approaches by predicting latent targets rather than pixels.

Parameter-efficient fine-tuning. Full fine-tuning a 307M-parameter encoder is costly on a small task-specific subset. PEFT methods reduce the trainable parameter count, with adapter layers [Houlsby et al., 2019], prefix tuning [Li and Liang, 2021], and LoRA [Hu et al., 2022] being common examples. LoRA keeps the original weights fixed and learns a low-rank residual update. Our focus is empirical: whether this low-rank update is sufficient for a feature-prediction video encoder, and how rank and layer placement affect anticipation.

Attentive probing. Frozen feature evaluation relies on probes to measure representation quality. We adopt the attentive probing design from V-JEPA 2 [Assran et al., 2025] and the prior V-JEPA work [Bardes et al., 2024] and train a shared transformer-based probe across all experiments. The probe serves as a fixed evaluation head; it is part of our setup, not a contribution.

3 Method

3.1 V-JEPA 2 Backbone

We use the V-JEPA 2 ViT-Large encoder `facebook/vjepa2-vitl-fpc64-256` [Assran et al., 2025]. The backbone has 307M parameters, 24 transformer layers, a 1024-dimensional embedding, and 16 attention heads. It uses three-dimensional rotary positional embeddings (RoPE) for spatiotemporal position encoding. V-JEPA 2 was pre-trained by Meta on approximately 22M videos using a joint-embedding feature-prediction objective. During pre-training, a context encoder predicts latent targets from an exponential-moving-average target encoder under high-ratio block masking. We adapt the released context encoder in all downstream experiments.

Table 1: Architecture and training configuration.

Component	Configuration
Backbone	V-JEPA 2 ViT-Large
Backbone size	307M parameters
Encoder	24 layers, 1024 dim., 16 heads
Position encoding	3D RoPE
LoRA targets	Query and value projections
LoRA ranks	4, 8, 16, 32
LoRA scaling	$\alpha = 2r$
Attentive probe	4 blocks, 3 query tokens, 67.4M params
Input	32 frames, 8 FPS, 256 ² px
Context / gap	4 s / 1 s
Optimizer	AdamW, batch size 8
Learning rate	10^{-4} (baseline) / 5×10^{-5} (LoRA)

3.2 LoRA Adaptation

For a frozen pre-trained projection matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA [Hu et al., 2022] learns a low-rank residual update:

$$W = W_0 + \Delta W, \quad \Delta W = BA, \quad (1)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are updated during training. The adapter output is scaled by α/r before addition, where we set $\alpha = 2r$ so that the effective scaling is constant across all rank configurations.

Using the Hugging Face PEFT library, we inject LoRA adapters into the query and value projections of self-attention. Our main variants apply adapters to all 24 encoder layers. We conduct a rank sensitivity study with $r \in \{4, 8, 16, 32\}$, corresponding to 0.39M, 0.79M, 1.57M, and 3.15M trainable adapter parameters. To study layer placement under a fixed budget, we also apply rank-16 adapters only to layers 18–23. This upper-layer configuration contains 0.39M adapter parameters, matching full-layer LoRA with $r = 4$.

3.3 Attentive Probe

All experiments use the same attentive probe, trained from scratch. The probe has four pre-LayerNorm transformer blocks, 16 attention heads, an MLP ratio of 4.0, and zero dropout. Three learnable cross-attention query tokens summarize the video representation for verb, noun, and action classification. The probe totals 67.4M trainable parameters. For the frozen baseline, only the probe is trained. For LoRA variants, the probe and low-rank adapter matrices are trained jointly while the original backbone weights remain frozen. Table 1 summarizes the full architecture and training configuration.

4 Experimental Setup

4.1 Dataset and Subset Construction

We train and evaluate on a 22-participant subset of EPIC-KITCHENS-100 [Damen et al., 2022], comprising approximately 56,311 training clips and 6,720 validation clips from the official train and validation CSVs. The full dataset spans 37 participants; our subset was determined only by frame-storage capacity, and all experiments use the same subset. We therefore report internally controlled comparisons and discuss full-dataset V-JEPA results only as context in Section 6.

Frames are pre-extracted JPEG images obtained from the official EPIC-KITCHENS-100 distribution hosted by the University of Bristol [Damen et al., 2022]. The probe predicts the full training vocabulary: 97 verbs, 300 nouns, and 3,568 verb–noun actions. Mean-class recall is computed over validation-present classes, including 895 actions in our subset.

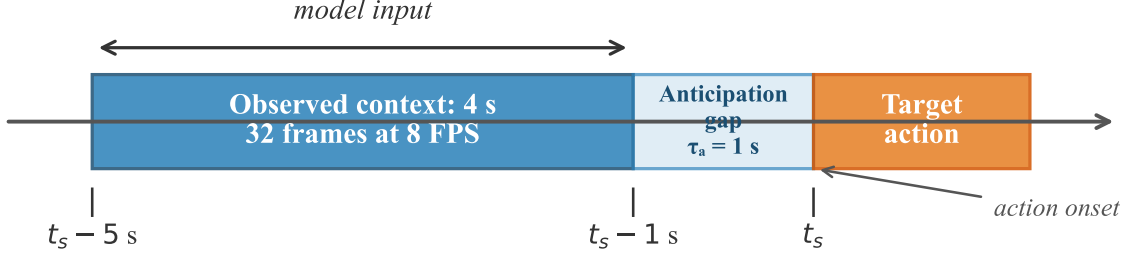


Figure 1: Temporal anticipation protocol. The model receives a four-second context window ending one second before the target action begins.

4.2 Temporal Anticipation Protocol

For an action starting at time t_s , the observation ends one second before its onset:

$$t_{\text{end}} = t_s - \tau_a, \quad \tau_a = 1.0\text{ s}. \quad (2)$$

We sample a four-second context window ending at t_{end} , as illustrated in Figure 1. Each input therefore contains 32 frames at 8 FPS, linearly sampled from source video recorded at 60 FPS. The model sees none of the anticipated action.

The one-second gap follows the standard EPIC-KITCHENS-100 anticipation protocol [Damen et al., 2022]. The four-second, 8 FPS window gives V-JEPA 2 its expected 32-frame input, covers immediate pre-action context, and avoids near-duplicate frames from the 60 FPS source videos. These temporal settings were fixed before the LoRA sweep and held constant across all variants. Frames are processed at 256×256 resolution with center cropping and the default normalization from the Hugging Face `AutoVideoProcessor`. We use no additional data augmentation.

4.3 Loss Function

We optimize focal loss [Lin et al., 2017] independently for verbs, nouns, and actions. For ground-truth class probability p_t , focal loss is:

$$\mathcal{L}_{\text{focal}}(p_t) = -(1 - p_t)^\gamma \log(p_t), \quad (3)$$

with $\gamma = 2.0$. The complete objective gives equal weight to each prediction head:

$$\mathcal{L} = \mathcal{L}_{\text{verb}} + \mathcal{L}_{\text{noun}} + \mathcal{L}_{\text{action}}. \quad (4)$$

We use `ignore_index=-1` for unseen action classes.

4.4 Optimisation

We train with AdamW [Loshchilov and Hutter, 2019] using $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay 0.05. The baseline learning rate is 10^{-4} ; all LoRA variants use 5×10^{-5} . A linear warmup covers the first 5% of steps, followed by cosine decay to zero. We clip the gradient norm to 1.0 and train with a batch size of eight using `bfloat16` mixed precision.

Experiments run on two NVIDIA A100 80 GB GPUs with PyTorch 2.6 and PEFT 0.19.

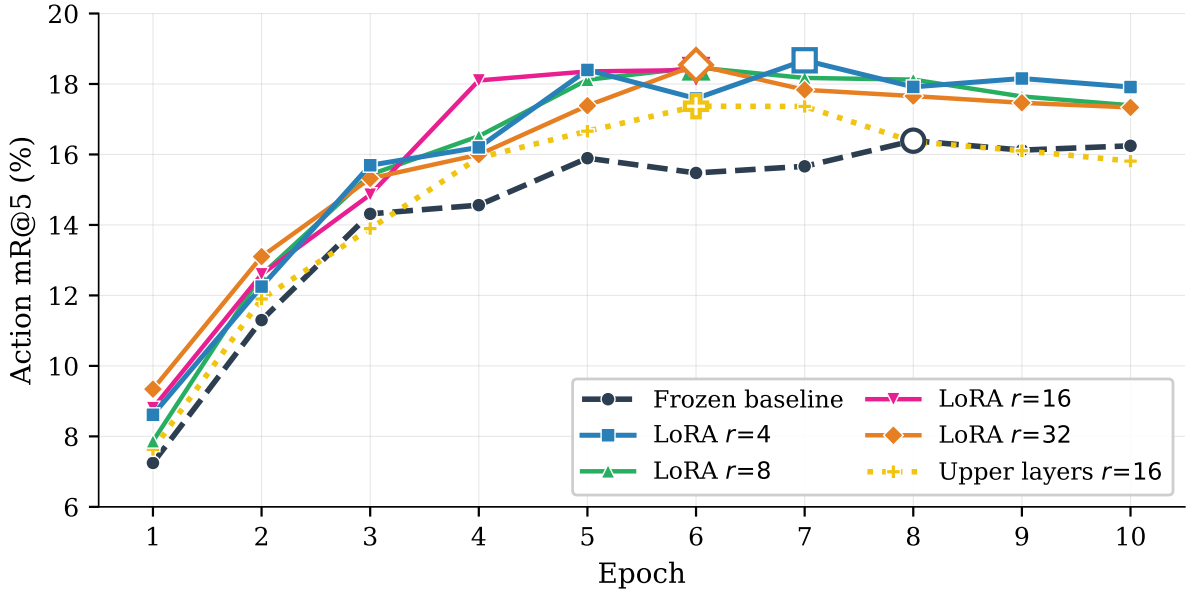


Figure 2: Validation action mR@5 over training epochs. Low-rank full-layer adaptation consistently improves on frozen probing. Increasing rank beyond $r = 4$ does not improve the best action score.

4.5 Evaluation Metrics

Our primary metric is mean-class Recall@5 (mR@5), following EPIC-KITCHENS evaluation practice. Mean-class recall suits the long-tailed label distribution because it weights each class equally instead of letting frequent actions dominate. We report verb, noun, and action mR@5 separately.

5 Results

5.1 Main Results

Table 2: Best validation results selected by action mean-class Recall@5 (mR@5). Adapter parameters exclude the shared 67.4M-parameter attentive probe. Bold denotes the best value in each metric column.

Configuration	Adapter Params.	Best Epoch	Verb mR@5	Noun mR@5	Action mR@5
Frozen baseline	0	8	27.50	29.84	16.39
LoRA $r = 4$	0.39M	7	28.79	32.64	18.67
LoRA $r = 8$	0.79M	6	26.41	33.26	18.47
LoRA $r = 16$	1.57M	7	26.46	32.02	18.44
LoRA $r = 32$	3.15M	6	27.04	33.13	18.54
LoRA upper ($r = 16$)	0.39M	6	25.67	31.51	17.37

Table 2 compares frozen probing against each LoRA configuration. All best-epoch selections are made by action mR@5, the primary evaluation metric. Every LoRA variant improves noun and action mR@5 over the frozen baseline. Full-layer LoRA with $r = 4$ achieves the best result, raising action mR@5 from 16.39% to 18.67% (+2.28 pp absolute, +13.9% relative) with 0.39M adapter parameters (0.13% of the backbone).

5.2 Rank Sensitivity

Figure 2 shows validation action mR@5 across epochs. The full-layer LoRA variants cluster tightly at their best checkpoints: 18.67% for $r = 4$, 18.47% for $r = 8$, 18.44% for $r = 16$, and 18.54% for $r = 32$. Increasing

rank gives no monotonic benefit. This suggests that small adapter matrices are sufficient for this downstream update. LoRA $r = 32$ peaks at epoch 6 and drops to 17.34% by epoch 10 while training loss keeps falling, indicating overfitting.

Action and noun mR@5 improve across all LoRA variants. Verb mR@5 is mixed: most variants score below the baseline (26.41–27.04 vs. 27.50), though LoRA $r = 4$ reaches 28.79. Verb-relevant motion features appear to be well captured by the frozen encoder; LoRA helps most with noun and action composition.

5.3 Matched-Budget Layer Placement

Full-layer LoRA with $r = 4$ and upper-layer-only LoRA with $r = 16$ each add 0.39M adapter parameters. Performance differs: distributing small adapters across all 24 layers yields 18.67% action mR@5, while adapting only layers 18–23 yields 17.37%. The 1.30 pp gap shows that broad layer coverage matters more than concentrating the same budget in the upper layers.

5.4 Efficiency

Table 3: Training efficiency over ten epochs. Hours are summed from epoch-level logs.

Configuration	Peak GB	Total Hours
Frozen baseline	13.80	21.99
LoRA $r = 4$	52.93	39.58
LoRA $r = 8$	53.24	34.84
LoRA $r = 16$	53.31	35.88
LoRA $r = 32$	53.17	46.96
LoRA upper	32.42	35.75

Table 3 clarifies that efficiency here means trainable parameter count, not lower compute or memory. Compared with frozen probing, LoRA raises peak memory from 13.8 GB to about 53 GB and total training time from 22.0 to 34.8–47.0 hours because gradients and activations must pass through adapted encoder layers. The upper-layer variant reduces peak memory to 32.4 GB by adapting only six layers. Thus LoRA’s benefit is a compact adapter and unchanged backbone weights, not faster or lighter training than frozen probing; the timing numbers are descriptive rather than a hardware benchmark.

6 Limitations

We train on a 22-participant subset of EPIC-KITCHENS-100, so the results reflect internal comparisons among adaptation strategies and cannot be compared directly to full-dataset benchmarks.

Relation to reported V-JEPA results. Our best action score (18.67% mR@5) is below the EPIC-KITCHENS-100 figures reported for V-JEPA 2 (39.7%) [Assran et al., 2025] and V-JEPA 2.1 (40.8%) [Mur-Labadia et al., 2026], but these are not like-for-like comparisons. We use 22 of 37 participants, compute mean-class recall over the 895 action classes present in our validation subset, and train a 0.39M-parameter LoRA module plus probe rather than reproducing the full benchmark recipe. The results should therefore be read as an internally controlled comparison of PEFT strategies, not as a re-benchmarking of V-JEPA.

Comparison to full fine-tuning. We do not evaluate full fine-tuning of V-JEPA 2, whose activation-memory cost exceeded our two-GPU budget. Future work should evaluate the full 37-participant split, compare against full fine-tuning, and measure variance across seeds and augmentation settings.

7 Conclusion

We applied parameter-efficient adaptation to V-JEPA 2 for egocentric action anticipation. Low-rank updates to the query and value projections improve action mR@5 over frozen probing across all tested ranks. Rank $r = 4$ performs best, adding 0.39M parameters and raising action mR@5 by 2.28 percentage points. Distributing adapters across the full encoder beats concentrating the same budget in the upper six layers. These results indicate that low-rank adaptation is viable for JEPA video representations and can steer V-JEPA 2 toward egocentric anticipation with few trainable parameters. The strongest gains occur on noun and action prediction, suggesting that the frozen feature-prediction representation already encodes useful pre-action structure and benefits from light task specialization.

Future work includes evaluation on the full 37-participant EPIC-KITCHENS-100 split, a controlled comparison against full fine-tuning, and joint adaptation of the predictor and encoder.

References

- [Assran et al., 2025] Assran, M., Bardes, A., Fan, D., Garrido, Q., Howes, R., et al. (2025). V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*.
- [Assran et al., 2023] Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. (2023). Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [Bardes et al., 2024] Bardes, A., Garrido, Q., Ponce, J., Chen, X., Rabbat, M., LeCun, Y., Assran, M., and Ballas, N. (2024). Revisiting feature prediction for learning visual representations from video. In *European Conference on Computer Vision*.
- [Chen et al., 2020] Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning*.
- [Damen et al., 2018] Damen, D., Doughty, H., Farinella, G. M., Fidler, S., Furnari, A., et al. (2018). Scaling egocentric vision: The EPIC-KITCHENS dataset. *European Conference on Computer Vision*.
- [Damen et al., 2022] Damen, D., Doughty, H., Farinella, G. M., Furnari, A., Ma, J., et al. (2022). Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100. *International Journal of Computer Vision*.
- [Furnari and Farinella, 2019] Furnari, A. and Farinella, G. M. (2019). What would you expect? anticipating egocentric actions with rolling-unrolling LSTMs and modality attention. In *International Conference on Computer Vision*.
- [He et al., 2022] He, K., Chen, X., Xie, S., Li, Y., Dollar, P., and Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [He et al., 2020] He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [Houlsby et al., 2019] Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. (2019). Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*.

- [Hu et al., 2022] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- [Kay et al., 2017] Kay, W., Carreira, J., Simonyan, K., Zhang, B., Hillier, C., et al. (2017). The Kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*.
- [Li and Liang, 2021] Li, X. L. and Liang, P. (2021). Prefix-tuning: Optimizing continuous prompts for generation. In *Annual Meeting of the Association for Computational Linguistics*.
- [Lin et al., 2017] Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollar, P. (2017). Focal loss for dense object detection. In *International Conference on Computer Vision*.
- [Loshchilov and Hutter, 2019] Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *International Conference on Learning Representations*.
- [Mur-Labadia et al., 2026] Mur-Labadia, L., Muckley, M., Bar, A., Assran, M., Sinha, K., Rabbat, M., LeCun, Y., Ballas, N., and Bardes, A. (2026). V-JEPA 2.1: Unlocking dense features in video self-supervised learning. *arXiv preprint arXiv:2603.14482*.
- [Radford et al., 2021] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., et al. (2021). Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*.
- [Tong et al., 2022] Tong, Z., Song, Y., Wang, J., and Wang, L. (2022). VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Advances in Neural Information Processing Systems*.